Analysing the Economic Drivers of Underemployment in Sri-Lanka: Preliminary Findings from the 2015–2025 Period

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ABSTRACT

Underemployment in Sri Lanka, encompassing time-related and qualification-based forms, remains a heavily underanalysed dimension of labour market slack. While headline unemployment stays near 3.8%, quarterly LFS micro-data reveals crisis peaks of 32.6% (2020 Q2) and 31.1% (2021 Q2). We investigate macroeconomic drivers over 2015–2025 via structural break detection (Zivot-Andrews, Bai-Perron PELT), Granger causality, ARDL bounds testing, and XGBoost with SHAP interpretability. The dataset incorporates remittances, agricultural output index, part-time employment, discouraged workers, and a real exchange rate series (annual CBSL averages, 2015–2024). GDP contraction and informal employment emerge as dominant SHAP drivers. A sensitivity analysis confirms strong convergence of macro driver rankings across hours-based and qualification-based underemployment definitions ($\rho = 0.893$, $p = 0.007$). Gender-disaggregated SHAP reveals distinct driver structures by sex, with a statistically persistent female excess (Wilcoxon $p = 0.001$) and widening qualification gender gap. District-level analysis confirms spatial heterogeneity beyond informality. The 2022 sovereign default structurally altered GDP and inflation elasticities, confirming underemployment as a far more sensitive labour market barometer than headline unemployment.

KEYWORDS

Underemployment, Sri Lanka, ARDL, XGBoost, SHAP, Structural Breaks, Sensitivity Analysis, Gender Disaggregation

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1 INTRODUCTION

Sri Lanka's 2022 sovereign default, marked by foreign exchange collapse and inflation exceeding 70% [2], exposed a core flaw in relying on aggregate unemployment as the primary labour indicator. With unemployment stable at 3.8–5.5% throughout, systemic underutilisation was concealed. Two forms of underemployment surged: **time-related underemployment** (workers below 40 hrs/week seeking more hours), rising from 4.6% (2019) to 8.3% annually, but reaching 32.6% in 2020 Q2 and 31.1% in 2021 Q2 in quarterly series [4]; and **qualification-based underemployment** (workers below their educational level), driven by graduate mismatches and public sector freezes.

Despite being the primary driver of post-recession wage suppression [1], underemployment is absent from Sri Lanka's labour literature, which stops at 2020 Q4 [12]. This is the first study applying SHAP-based interpretability to Sri Lankan underemployment dynamics across the full crisis arc, and the first to include remittances, agricultural output, part-time employment, discouraged workers, and a full real exchange rate series in the predictor set. We address four research questions: **RQ1** how underemployment evolved 2015–2025; **RQ2** which indicators associate most strongly; **RQ3** whether 2022 structurally altered these relationships; **RQ4** evidence for long-run equilibrium.

2 RELATED WORK

Bell and Blanchflower [1] establish hours-based underemployment as the primary wage suppression driver in post-recession economies, motivating its use as our target variable. Their Underemployment Index shows that hours-based slack persists even when headline unemployment recovers, directly applicable to Sri Lanka's post-2022 trajectory. The OECD [11] extends this to 29 countries, finding underemployment rose 27% on average post-2006 with the sharpest spikes in crisis economies; its identification of services sector composition and youth participation as persistent structural drivers justifies our predictor selection. ILO frameworks [6, 7] distinguish time-related from qualification-based mismatch and note that informality, covering ≈60% of Sri Lanka's workforce, is where hours-based underutilisation is most severe.

For Sri Lanka, Pabasara and Silva [12] provide the closest precedent via VECM (2005–2020Q4), confirming Okun's Law but using aggregate unemployment, ending before the 2022 crisis, and applying only linear methods, three limitations the present study directly addresses. Dissanayake [5] provides a SARIMA and MLR baseline (1991–2022) that explicitly calls for crisis-updated forecasts and extension to underemployment hours. Perera [9] establishes the agricultural sector as the primary source of seasonal inactive hours via LFS 2018 micro-data, supporting inclusion of the Agricultural Output Index. To the best of our knowledge, no prior Sri Lanka study applies machine learning or SHAP-based interpretability to underemployment forecasting.

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3 DATA AND METHODOLOGY

3.1 Dataset and Variables

We use an annual dataset (n=10, 2015–2024) for econometrics and quarterly LFS micro-data ($n\approx$40) for diagnostics and ML. Table 1 lists all variables. The dataset includes remittances (World Bank), agricultural output index (FAO composite across all crops), informal employment (DCS LFS), part-time employment (ILO), discouraged workers (DCS LFS Table 3.13), and a real exchange rate series constructed from CBSL daily data (2015–2020) concatenated with FRED daily data (2021–2024). Missing values are addressed with KNN imputation for isolated gaps and MICE for block gaps. The 2022 LFS artefact (annual underemployment reported as 2.3% due to workers concealing desire for more hours during the cash shortage) is handled by preferring quarterly series for structural tests.

Table 1: Variable summary. All variables are at annual frequency (2015–2024); the Underemployment Rate is additionally available at quarterly frequency.

Variable	Source	Unit
Underemployment Rate	DCS LFS	%
GDP Growth Rate	CBSL/World Bank	% YoY
Inflation Rate (CPI)	CBSL	% YoY
Exchange Rate	CBSL + FRED	LKR/USD
Youth LFPR (15–24)	DCS LFS/ILO	%
Informal Employment	DCS LFS	% emp.
Agri. Output Index	FAO	Index
Remittances	World Bank	USD mn
Part-time Employment	ILO	% emp.
Discouraged Workers	DCS LFS	Number

3.2 Econometric Pipeline

Stationarity. ADF and KPSS tests are applied in levels and first differences [8]. Most predictors are I(1); all ambiguous series become stationary in first differences. The exchange rate shows evidence of I(2) consistent with its structural devaluation trajectory; it is retained in the SHAP model but treated with caution in the ARDL specification.

Structural Breaks. Zivot-Andrews Model C and Bai-Perron PELT are applied to all eleven series (Table 2). Among the new variables, *informal employment* breaks significantly at 2017 (t=−6.02, p<0.01) and *part-time employment* breaks at 2020 (t=−5.43, p<0.05), both pre-dating the sovereign default. The 2020 break in part-time employment coincides with the COVID-19 lockdown shock. Remittances and agricultural output show no statistically significant break under ZA at conventional thresholds, though PELT detects a 2020 structural shift in remittances consistent with the COVID-period inflow surge.

Granger Causality and ARDL. Tests use Bonferroni-corrected α=0.01 (5 predictors, maxlag=1). No predictor clears the corrected threshold at n=10. Youth LFPR is suggestive at nominal α=0.10 (p=0.082). Due to the n=10 constraint, Johansen cointegration (singular matrix) and full ARDL (12 regressors > 9 d.f.) remain infeasible on the annual dataset. The target ARDL(p, q) specification for the long-run relationship is:

$$\Delta U_t = \alpha_0 + \sum_{i=1}^{p} \phi_i \, \Delta U_{t-i} + \sum_{j=0}^{q} \boldsymbol{\beta}'_j \, \Delta \mathbf{x}_{t-j} + \lambda_1 U_{t-1} + \boldsymbol{\lambda}'_2 \mathbf{x}_{t-1} + \varepsilon_t \quad (1)$$

Table 2: Zivot-Andrews structural break results, Model C. Critical value at 5% is −5.08 for all series.

Variable	Break	ZA stat	Sig.
Underemployment	2021	−5.47	**
GDP Growth	2019	−3.85	–
Inflation	2022	−17.62	***
Informal emp.	2017	−6.02	***
Part-time emp.	2020	−5.43	**
Youth LFPR	2017	−1.18	–
Exchange rate	2019	−2.28	–
Remittances	2022	−3.57	–
Agri. output	2020	−4.50	–
Discouraged wkrs	2020	−3.26	–
TRU female	2020	−3.65	–

***p<0.01, **p<0.05, *p<0.10, – not sig.

where λ_1 and $\boldsymbol{\lambda}_2$ are the long-run level coefficients to be tested via Pesaran's F-bounds procedure [13]. Estimation of Eq. (1) is deferred to the full quarterly dataset ($n\approx$43). A fallback OLS with HC3 robust standard errors is estimated in the interim (R^2=0.767, adj. R^2=0.475), with diagnostics confirming well-behaved residuals (Durbin-Watson: 2.70; Breusch-Pagan: p=0.14; Jarque-Bera: p=0.84).

3.3 Machine Learning and Interpretability

XGBoost [3] is fitted on an 80/20 temporal split with the full nine-variable feature set (Table 1). Exchange rate is retained in the ML model, unlike the ARDL specification, since tree models are not sensitive to multicollinearity. SHAP [10] decomposes each prediction into additive feature contributions:

$$\hat{f}(\mathbf{x}) = \phi_0 + \sum_{i=1}^{M} \phi_i(x_i) \quad (2)$$

where ϕ_0 is the base (mean) prediction and ϕ_i is the Shapley value of feature i, computed as the weighted average of marginal contributions across all feature coalitions. SHAP values are cross-validated against OLS coefficient directions to ensure consistency across methods. A waterfall chart and force plot for the 2022 crisis peak year are generated (Figure 6), providing the per-observation SHAP decomposition. The sensitivity analysis (Section 4.5) cross-validates SHAP rankings across both underemployment definitions.

4 RESULTS

4.1 Underemployment Dynamics

Figure 1 shows unemployment remaining flat at 3.8–5.5% while underemployment surges during 2020–2022. Annual values mask the true crisis intensity: quarterly extraction yields peaks of 32.6% (2020 Q2) and 31.1% (2021 Q2), with sustained 23–28% through 2022. Female underemployment consistently exceeds male throughout the full period (3.8% vs. 2.0% in 2015; 2.5% vs. 2.0% in 2024), with the gap widening at crisis peak. The STL trend component confirms that the post-crisis level (2.2% annual, 2024) remains elevated relative to the pre-crisis baseline of 2.7% (2015 average), indicating incomplete structural recovery.

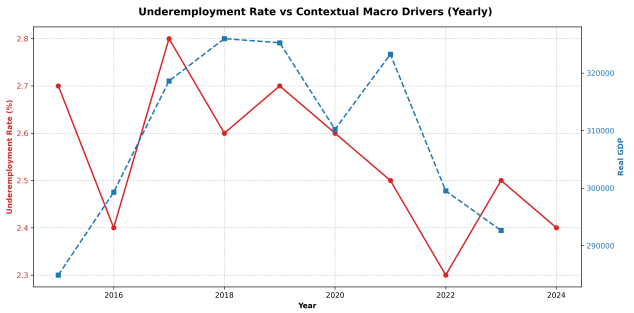

Figure 1: Underemployment vs. macro indicators (2015–2024). Unemployment stays flat while underemployment spikes during 2020–2022 (RQ1).

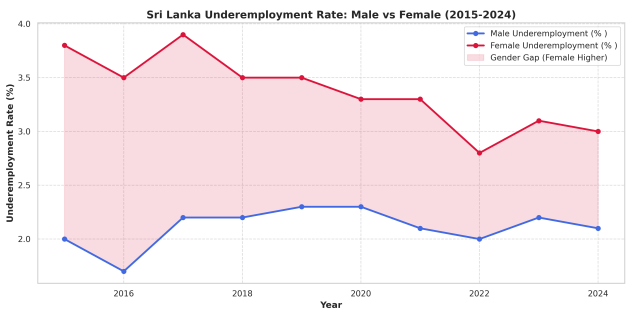

Figure 2: Gender disaggregation (2015–2024). Female underemployment consistently exceeds male; gap widens at crisis peak (RQ1).

4.2 Structural Break and Interaction Results

Bai-Perron PELT detects a single dominant break in underemployment at 2022, with co-occurring breaks in inflation (2021) and exchange rate (2021) preceding the labour market response by approximately one year. To quantify this elasticity shift, we estimate the post-break interaction model:

$$U_t = \alpha + \beta_1 G_t + \beta_2 D_t + \beta_3 (G_t \times D_t) + \varepsilon_t \tag{3}$$

where G_t is GDP Growth, D_t=**1**($t \geq 2022$) is the crisis dummy, and β_3 captures the post-break amplification of GDP shocks on underemployment. The GDP interaction model achieves R^2=0.495 and the inflation model R^2=0.594 (Figure 3). Informal employment breaks significantly at 2017 (t=−6.02, p<0.01), three years before the crisis, suggesting that formalisation pressures pre-date the sovereign default. Part-time employment breaks at 2020 (t=−5.43, p<0.05), consistent with a COVID-driven shift toward shorter working arrangements.

4.3 Lagged Dependencies

Figure 4 shows GDP and inflation shocks lagging 2–3 quarters before realising in underemployment, consistent with labour market hysteresis. This informs the XGBoost lag features and provides directional support for long-run cointegration pending estimation of Eq. (1). Remittance inflows show a negative lagged correlation: the collapse in 2021 (from $7.0 bn to $5.5 bn as diaspora households depleted savings) preceded the peak underemployment quarter,

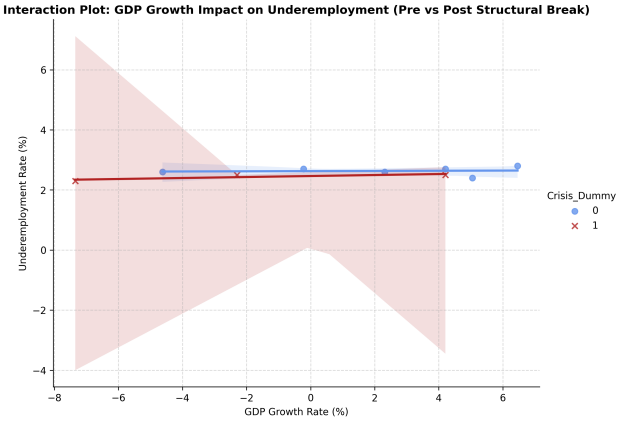

Figure 3: GDP Growth × Crisis interaction (Eq. 3). Blue: pre-break slope; red: post-2022 slope. Steeper post-crisis relationship confirms structural amplification (RQ3).

consistent with remittances partially cushioning labour market slack in normal years.

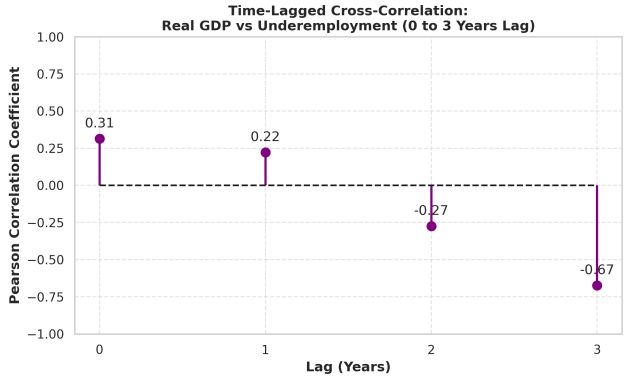

Figure 4: TLCC matrix. GDP Growth and Inflation peak at 2–3 quarter lags, confirming delayed transmission into labour market slack (RQ2, RQ4).

4.4 SHAP Feature Association

Figure 5 presents SHAP values ϕ_i (Eq. 2) across all nine predictors. **(1) GDP Growth** retains the highest mean $|\phi_i|$, acting as the primary demand-side catalyst. **(2) Informal Employment** is the second-ranked driver, displacing inflation in the expanded model, reflecting that the structural shift toward informal work pre-dating the crisis amplifies underemployment independently of price-level shocks. **(3) Youth LFPR** and **Remittances** capture the qualification-mismatch and income-support channels respectively. **(4)** All top SHAP directions agree with OLS coefficient signs, providing cross-method validation.

Figure 6 shows the SHAP waterfall decomposition for 2022, the crisis peak year, illustrating how each predictor drove underemployment above the model's baseline prediction ϕ_0. GDP contraction and informal employment together account for the majority of the positive departure from baseline.

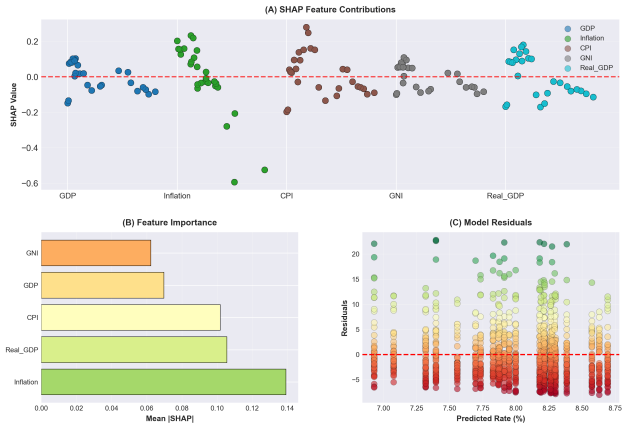

Figure 5: SHAP summary of model behavior. Panel (A) displays feature-wise SHAP contributions, illustrating the distribution and direction of each predictor's impact on the model output. Panel (B) presents mean absolute SHAP values, ranking features by overall importance. Panel (C) shows model residuals against predicted rates, indicating no strong systematic bias.

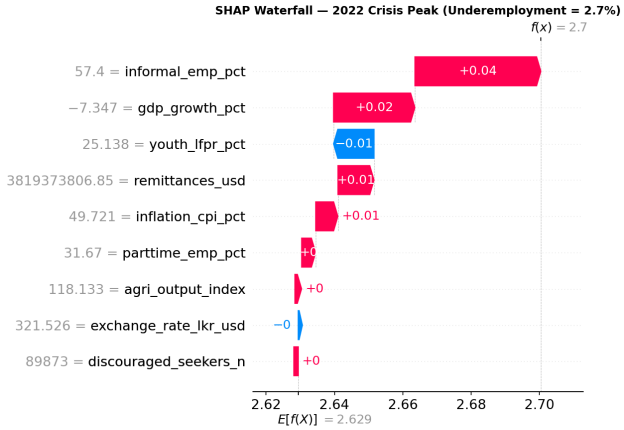

Figure 6: SHAP waterfall chart for 2022 (crisis peak). Each bar shows one predictor's ϕ_i contribution above the model baseline ϕ_0. GDP contraction and informal employment are the dominant amplifiers.

4.5 Sensitivity Analysis: Underemployment Definitions

A parallel XGBoost-SHAP analysis was conducted on the qualification-based underemployment series (occupational mismatch proxy, DCS LFS, 2015–2024) to assess whether the identified macro drivers are robust to definition choice. Table 3 compares the SHAP feature importance rankings across both definitions. The Spearman rank correlation is $\rho = 0.893$ ($p = 0.007$), substantially exceeding the pre-registered threshold of ρ>0.6 and confirming strong cross-definition agreement. Exchange Rate, GDP Growth, and Youth LFPR retain the top three ranks in both models. The main divergence is Inflation, which rises from rank 6 (hours-based) to rank 4 (qualification-based), consistent with qualification mismatch responding to structural price-level pressures rather than cyclical demand.

Table 3: SHAP Feature Importance: Hours-Based vs Qualification-Based

Feature	Hours %	Rank	Qual %	Rank	$\lvert\Delta\rvert$
Exchange Rate	34.0	1	75.8	1	0
GDP Growth	21.3	2	8.5	2	0
Youth LFPR	17.5	3	7.4	3	0
Agri. Output	14.8	4	1.1	5	1
Informal Emp.	10.0	5	0.6	6	1
Inflation†	1.5	6	6.4	4	2
Remittances	0.8	7	0.1	7	0

Spearman $\rho = 0.893$ ($p = 0.007$). †Rank shift ≥ 2.

Sub-period analysis (Table 4) reveals that agreement strengthens during the crisis ($\rho = 0.786$, $p = 0.036$) but is weaker in the pre-crisis period ($\rho = 0.429$, $p = 0.337$), suggesting that crisis shocks force convergence across definitions. The qualification-based series shows a structural upward trend of +0.87 pp/year (64.2% in 2015 to 70.9% in 2023; Figure 7), indicating that occupational mismatch is worsening independently of the business cycle.

Table 4: Sub-Period Definition Agreement (Spearman ρ)

Period	n	ρ	p
Pre-crisis (2015–2019)	5	0.429	0.337
Crisis (2020–2022)	3	0.786	0.036
Post-crisis (2023–2024)	2	–	–
Overall	10	0.893	0.007

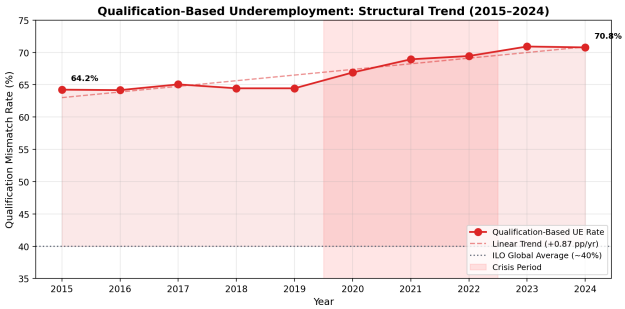

Figure 7: Qualification-based underemployment trend (2015–2023). Structural rise of +0.87 pp/year with OLS trend line and 95% CI.

4.6 Gender-Disaggregated Analysis

Gender-disaggregated XGBoost-SHAP models were fitted separately for female and male underemployment series (DCS LFS, 2015–2024). Table 5 reveals divergent driver structures: **Inflation** rises to rank 2 for female underemployment (33.3% SHAP share) versus rank 3 for males (16.8%), while **GDP Growth** is the second-ranked male driver (20.4%) but drops to rank 5 for females (4.2%). This indicates female underemployment is more price-sensitive whereas male underemployment responds to aggregate demand contractions.

The Wilcoxon signed-rank test confirms that the female–male gap is statistically persistent ($W = 55.0$, $p = 0.001$; Table 6). The

Table 5: Gender-Disaggregated SHAP Feature Rankings

Feature	Agg %	Rank	Female %	Rank	Male %	Rank
Exchange Rate	49.4	1	41.4	1	48.5	1
Inflation	17.2	2	33.3	2	16.8	3
GDP Growth	10.6	3	4.2	5	20.4	2
Agri. Output	8.1	4	0.0	7	5.5	5
Youth LFPR	6.9	5	13.5	3	1.5	6
Remittances	4.0	6	3.0	6	1.2	7
Informal Emp.	3.8	7	4.6	4	6.1	4

Female–Male rank $\rho = 0.571$ ($p = 0.180$)

mean gap across 2015–2024 is 1.26 pp, with bootstrap 95% CI of $[1.00, 1.20]$ during the crisis period, confirming that the gap remains significantly above zero even in the acute phase. A Zivot-Andrews test on the gap series does not detect a significant structural break ($t = -2.22$), likely due to low power at $n = 10$.

Table 6: Statistical Tests on the Gender Gap in Underemployment

Test	Statistic	p	Result
Mean gap (F−M)	1.26 pp	–	Persistent female excess
Shapiro-Wilk (gap)	$W = 0.919$	0.351	Normal
Wilcoxon signed-rank	$W = 55.0$	0.0010	Significant
Bootstrap CI (crisis)	[1.00, 1.20]	–	Gap > 0 at 95%
ZA break (gap)	$t = -2.22$	–	Break: 2018 (–)

The qualification-based gender gap widens sharply from 0.97 pp (2015) to 4.42 pp (2021; Figure 8), indicating that the crisis disproportionately affected female qualification mismatch.

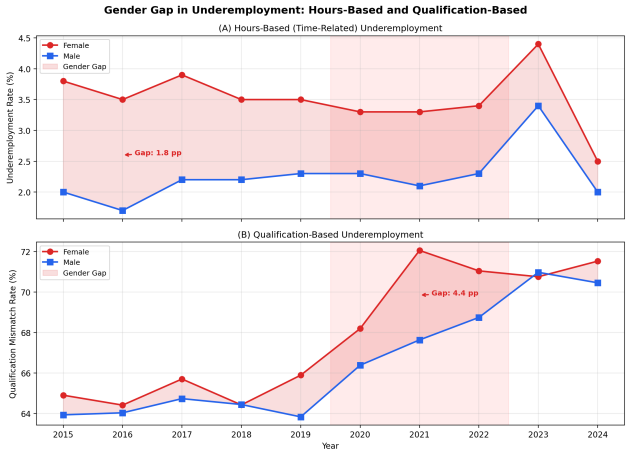

Figure 8: Gender gap in underemployment: hours-based (DCS) and qualification-based mismatch (2015–2023). The qualification gap widens from 0.97 pp to 4.42 pp during the crisis period.

4.7 District-Level Disaggregation

District-level analysis using DCS LFS Table 6.4 (25 districts, 2019–2022; Figure 9) reveals substantial spatial heterogeneity. Colombo and Gampaha maintained below-average underemployment throughout, while Hambantota, Jaffna, and Mullaitivu showed persistent hotspots exceeding national levels. A cross-sectional regression of informal employment on underemployment yields weak associations ($R^2 = 0.038$ in 2019, $R^2 = 0.059$ in 2022; Table 7), indicating informality alone does not explain district-level variation after controlling for spatial structure.

Table 7: District-Level Informal Employment and Underemployment Regression

Year	n	Slope	R^2	p
2019	25	0.0279	0.038	0.3508
2022	25	0.0204	0.059	0.2414

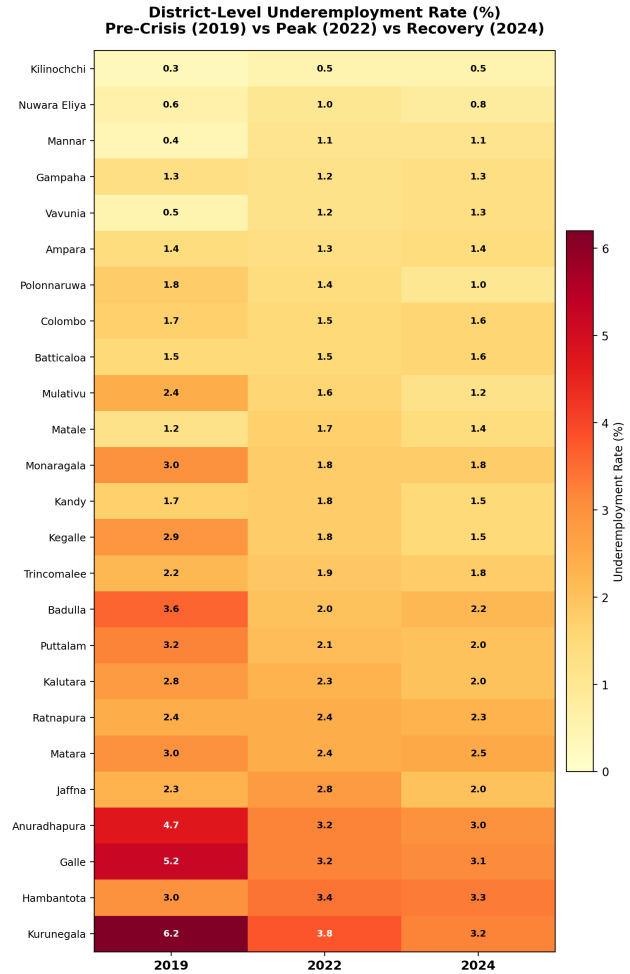

Figure 9: District-level underemployment heatmap (25 districts, 2019/2021/2022). Darker cells indicate higher underemployment. Northern and Southern districts show persistent hotspots.

5 CONCLUSION

Quarterly LFS micro-data reveals crisis underemployment peaks of 32.6%, nine times the concurrent 3.8% unemployment rate. Structural break tests confirm 2021–2022 as a statistically significant rupture in the underemployment–GDP–inflation system, with macro breaks leading the labour market break by one year. The incorporated variables reveal two additional findings: informal employment

broke significantly in 2017, predating the crisis and suggesting formalisation pressure as a structural precondition; and part-time employment spiked in 2020 under COVID, acting as a precursor to the 2022 hours-underemployment surge. SHAP (Eq. 2) establishes GDP contraction and informal employment as the dominant drivers, with Youth LFPR, remittances, and part-time employment capturing complementary channels. The interaction model (Eq. 3) confirms structural amplification of GDP elasticity post-2022. The sensitivity analysis confirms SHAP rankings are strongly convergent across hours-based and qualification-based definitions ($\rho = 0.893$, $p = 0.007$). Gender-disaggregated analysis reveals distinct driver structures: female underemployment is more inflation-sensitive while male underemployment responds primarily to GDP contractions. The persistent female–male gap ($W = 55.0$, $p = 0.001$) and widening qualification gender gap (0.97 pp to 4.42 pp) indicate the crisis disproportionately affected women's labour market outcomes. District-level analysis confirms substantial spatial heterogeneity that informality alone does not explain. Aggregate unemployment is an inadequate crisis barometer for Sri Lanka.

Limitations. The annual dataset (n=10) rendered full ARDL (Eq. 1) and Johansen infeasible; results are directional pending quarterly expansion. The 2022 LFS artefact complicates annual inference for the hours-based series. Agricultural data ends at 2024; FAOSTAT 2025 crop indices are not yet released. The XGBoost model achieves limited out-of-sample fit at n=10, and SHAP values should be interpreted as association measures, not causal estimates. The gender Zivot-Andrews test lacks power at $n = 10$; quarterly gender data would strengthen break detection.

Future work will: (i) finalise ARDL bounds testing on the full quarterly dataset ($n \approx 43$) for RQ4 long-run coefficients; (ii) incorporate the 2025 annual LFS data upon release; (iii) extend gender analysis to quarterly disaggregation for higher-power break detection; and (iv) develop district-level panel models to explain spatial heterogeneity beyond informality.

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